

MINI PROJECT

Employee_Information

1. Create the following tables.

EMP_Details

EMP_Skills

SKILLS_Details

LODGE_Details

```
SQL> CREATE TABLE LODGE_DETAILS
2 (
3     LODGE_NAME VARCHAR2(30) PRIMARY KEY,
4     LODGE_MANAGER VARCHAR2(30) NOT NULL,
5     LODGE_ADDRESS VARCHAR2(30)
6 );
```

Table created.

```
SQL> CREATE TABLE EMP_DETAILS
2 (
3     EMPNO INTEGER PRIMARY KEY,
4     FIRST_NAME VARCHAR2(30) NOT NULL,
5     LAST_NAME VARCHAR2(30) NOT NULL,
6     LODGE_NAME VARCHAR2(30),
7     CONSTRAINT FK_EMP_LODGE
8     FOREIGN KEY (LODGE_NAME)
9     REFERENCES LODGE_DETAILS(LODGE_NAME)
10 );
```

Table created.

```
SQL> CREATE TABLE SKILL_DETAILS
2 (
3     SKILL VARCHAR2(30) PRIMARY KEY,
4     SKILL_DESC VARCHAR2(30)
5 );
```

Table created.

```
SQL> CREATE TABLE EMP_SKILL
2 (
3     EMPNO INTEGER,
4     SKILL VARCHAR2(30),
5     GRADE NUMBER(10),
6     CONSTRAINT PK_EMP_SKILL
7     PRIMARY KEY (EMPNO, SKILL),
8     CONSTRAINT FK_EMP
9     FOREIGN KEY (EMPNO)
10     REFERENCES EMP_DETAILS(EMPNO),
11     CONSTRAINT FK_SKILL
12     FOREIGN KEY (SKILL)
13     REFERENCES SKILL_DETAILS(SKILL)
14 );
```

Table created.

2. Insert data into the following tables

```
SQL> INSERT INTO LODGE_DETAILS VALUES
2 ('Jessy Lodge', 'Rajan', 'Rakesh Nagar');
```

1 row created.

```
SQL>
SQL> INSERT INTO LODGE_DETAILS VALUES
2 ('Rajesh Lodge', 'Rajani', 'Anna Nagar');
```

1 row created.

```
SQL>
SQL> INSERT INTO LODGE_DETAILS VALUES
2 ('Gooday Lodge', 'John', 'Sea Road');
```

1 row created.

```
SQL> INSERT INTO EMP_DETAILS VALUES
  2 (101,'James','Jackson','Rajesh Lodge');

1 row created.

SQL>
SQL> INSERT INTO EMP_DETAILS VALUES
  2 (102,'Kalpash','Raj','Gooday Lodge');

1 row created.

SQL>
SQL> INSERT INTO EMP_DETAILS VALUES
  2 (103,'Jasmine','Joy','Jessy Lodge');

1 row created.
```

```
SQL> INSERT INTO SKILL_DETAILS VALUES
  2 ('Oracle SQL','Oracle Corporation');

1 row created.

SQL>
SQL> INSERT INTO SKILL_DETAILS VALUES
  2 ('Dot Net','Microsoft');

1 row created.

SQL>
SQL> INSERT INTO SKILL_DETAILS VALUES
  2 ('Sybase','SAP Corporation');

1 row created.
```

```
SQL> INSERT INTO EMP_DETAILS (EMPNO, FIRST_NAME, LAST_NAME, LODGE_NAME)
  2 VALUES (101, 'James', 'Jackson', 'Rajesh Lodge');

1 row created.

SQL> INSERT INTO EMP_DETAILS VALUES
  2 (102,'Kalpash','Raj','Gooday Lodge');

1 row created.

SQL>
SQL> INSERT INTO EMP_DETAILS VALUES
  2 (103,'Jasmine','Joy','Jessy Lodge');

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

3.Now the Employee 101 has been change his lodge to Gooday Lodge.

```
SQL> UPDATE EMP_DETAILS
  2 SET LODGE_NAME='Gooday Lodge'
  3 WHERE EMPNO=101;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.
```

4. Employee 103 has resigned from the company and vacated the lodge so delete his details from the respective table.

```
SQL> DELETE FROM EMP_SKILL
  2 WHERE EMPNO=103;

0 rows deleted.

SQL>
SQL> DELETE FROM EMP_DETAILS
  2 WHERE EMPNO=103;

1 row deleted.

SQL>
SQL> COMMIT;

Commit complete.

SQL> |
```

5. The HR Department want to create reports based on the following requirements.

a. Write query that will display the employee with skill Sybase. Pass Sybase in any case Use Function.

```
SQL> SELECT E.*
  2 FROM EMP_DETAILS E
  3 JOIN EMP_SKILL S
  4 ON E.EMPNO=S.EMPNO
  5 WHERE UPPER(S.SKILL)=UPPER('Sybase');

EMPNO FIRST_NAME LAST_NAME
-----
101 James Jackson
Gooday Lodge
```

b. Create a report that will display the First_Name, Sills and Lodge Manger

```
SQL> SELECT E.FIRST_NAME,
  2 S.SKILL,
  3 L.LODGE_MANAGER
  4 FROM EMP_DETAILS E
  5 JOIN EMP_SKILL S
  6 ON E.EMPNO = S.EMPNO
  7 JOIN LODGE_DETAILS L
  8 ON E.LODGE_NAME = L.LODGE_NAME;

FIRST_NAME SKILL
-----
James Sybase
John
Kalpash Oracle SQL
John
```

c. Create a report that will that display the number of skill that employee 101 has.

```
SQL> SELECT EMPNO,
  2 COUNT(SKILL) AS NUMBER_OF_SKILLS
  3 FROM EMP_SKILL
  4 WHERE EMPNO = 101
  5 GROUP BY EMPNO;

EMPNO NUMBER_OF_SKILLS
-----
101 1
```

d. Display the employee First_Name who has the Skills Oracle SQL.

```
SQL> SELECT E.FIRST_NAME
2 FROM EMP_DETAILS E
3 JOIN EMP_SKILL S
4 ON E.EMPNO = S.EMPNO
5 WHERE S.SKILL = 'Oracle SQL';
```

```
FIRST_NAME
-----
Kalpash
```

6.Create a View EMP_VIEW so that users can use it that contains columns EMPNO,FIRST_NAME,SKILLS,LODGE_MANAGER.

```
SQL> CREATE VIEW EMP_VIEW AS
2 SELECT E.EMPNO,
3 E.FIRST_NAME,
4 S.SKILL AS SKILLS,
5 L.LODGE_MANAGER
6 FROM EMP_DETAILS E
7 JOIN EMP_SKILL S
8 ON E.EMPNO = S.EMPNO
9 JOIN LODGE_DETAILS L
10 ON E.LODGE_NAME = L.LODGE_NAME;
```

View created.

7.Create A Synonym EMPV for the View EMP_VIEW

```
SQL> CREATE SYNONYM EMPV FOR EMP_VIEW;
```

Synonym created.

```
SQL> SELECT * FROM EMPV;
```

	EMPNO	FIRST_NAME	SKILLS
LODGE_MANAGER			
John	101	James	Sybase
John	102	Kalpash	Oracle SQL

```
SQL> SELECT * FROM USER_SYNONYMS;
```

SYNONYM_NAME
TABLE_OWNER
TABLE_NAME
DB_LINK
ORIGIN_CON_ID
DATA_PUMP_DIR_OBJS
SYS
LOADER_DIR_OBJS
SYNONYM_NAME
TABLE_OWNER
TABLE_NAME
DB_LINK

8. Create Index on Skill column in EMP_Skills table

```
SQL> CREATE INDEX IDX_EMP_SKILL
2 ON EMP_SKILL(SKILL);

Index created.

SQL> SELECT INDEX_NAME, TABLE_NAME, COLUMN_NAME
2 FROM USER_IND_COLUMNS
3 WHERE TABLE_NAME = 'EMP_SKILL';

INDEX_NAME
TABLE_NAME
COLUMN_NAME
-----
PN_EMP_SKILL
EMP_SKILL
EMPNO

PN_EMP_SKILL
EMP_SKILL
SKILL

INDEX_NAME
TABLE_NAME
COLUMN_NAME
-----

IDX_EMP_SKILL
EMP_SKILL
SKILL

SQL> SELECT *
2 FROM EMP_SKILL
3 WHERE SKILL = 'Sybase';

EMPNO SKILL GRADE
```

```
SQL> SELECT *
2 FROM EMP_SKILL
3 WHERE SKILL = 'Sybase';

EMPNO SKILL GRADE
-----
101 Sybase

SQL> |
```